Q1. $x_1 = (1, i, 0),\ x_2 = (1, 0, i),\ x_3 = (0, 0, 1)$

Sol: $c_1x_1 + c_2x_2 + c_3x_3 = 0$

$y_1 = x_1$

$z_1 = \frac{y_1}{\|y_1\|},\ \|y_1\| = \sqrt{|1|^2 + |i|^2 + |0|^2} = \sqrt{1+1} = \sqrt{2} \quad = \frac{1}{\sqrt{2}}(1, i, 0)$

$y_2 = x_2 - \text{proj}_{z_1}(x_2),\ \text{proj}_{z_1}(x_2) = \frac{\langle x_2, z_1\rangle}{\langle z_1, z_1\rangle} z_1$

$\langle x_2, z_1\rangle = \frac{1}{\sqrt{2}}(1\cdot 1 + 0\cdot i + i\cdot 0) = \frac{1}{\sqrt{2}};\ \text{proj}_{z_1}(x_2) = \frac{\frac{1}{\sqrt{2}}}{1}\cdot\frac{1}{\sqrt{2}}(1, i, 0) = \frac{1}{2}(1, i, 0)$

$y_2 = x_2 - \text{proj}_{z_1}(x_2) = (1, 0, i) - \frac{1}{2}(1, i, 0) = \left(\frac{1}{2}, \frac{1}{2}, i\right)$

$\|y_2\| = \sqrt{|\frac{1}{2}|^2 + |-\frac{1}{2}|^2 + |1|^2} = \frac{\sqrt{3}}{2};\ z_2 = \frac{1}{\|y_2\|} y_2 = \frac{1}{\sqrt{3}}\left(\frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}}, 1\right)$

$y_3 = x_3 - \text{proj}_{z_1}(x_3) - \text{proj}_{z_2}(x_3)$

$\langle x_3, z_1\rangle = \frac{1}{\sqrt{2}}(0\cdot 1 + 0\cdot i + 1\cdot 0) = 0,\quad \text{proj}_{z_1}(x_3) = 0$

$\langle x_3, z_2\rangle = \frac{1}{\sqrt{3}};\ \text{proj}_{z_2}(x_3) = \frac{1}{\sqrt{3}} z_2 = \frac{1}{3}\left(\frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}}, 1\right)$

$y_3 = \left(-\frac{1}{3\sqrt{2}}, \frac{i}{3\sqrt{2}}, \frac{2}{3}\right)$

$\|y_3\| = \sqrt{\frac{1}{18} + \frac{1}{18} + \frac{4}{9}} = 1;\ z_3 = y_3 = \left(-\frac{1}{3\sqrt{2}}, \frac{i}{3\sqrt{2}}, \frac{2}{3}\right)$

Finally, $z_1 = \frac{1}{\sqrt{2}}(1, i, 0),\ z_2 = \frac{1}{\sqrt{3}}\left(\frac{1}{\sqrt{2}}, -\frac{i}{\sqrt{2}}, 1\right),\ z_3 = \left(-\frac{1}{3\sqrt{2}}, \frac{i}{3\sqrt{2}}, \frac{2}{3}\right)$

Q2 $A = \begin{bmatrix} -2 & 1 & 1 \\ 0 & 3 & 0 \\ -5 & 1 & 4 \end{bmatrix} \qquad f(A) = 6A^3 - 17A^2 + A - 3I$

$\det(A - \lambda I) = 0$

$p(\lambda) = \det\begin{bmatrix} -2-\lambda & 1 & 1 \\ 0 & 3-\lambda & 0 \\ -5 & 1 & 4-\lambda \end{bmatrix}$

$= (-2)(12 - 7\lambda + \lambda^2) - \lambda(12 - 7\lambda + \lambda^2) + 15 - 5\lambda$

$\lambda = -2,\ \lambda = 3,\ \lambda = 4$

$m_A(\lambda) = (\lambda + 2)(\lambda - 3)(\lambda - 4)$

$m_A(A) = (A + 2I)(A - 3I)(A - 4I) = 0$

$A^2 = A\cdot A = \begin{bmatrix} -4 & 0 & -6 \\ 0 & 9 & 0 \\ -30 & 9 & 21 \end{bmatrix}$

$A^3 = A^2\cdot A = \begin{bmatrix} -8 & 9 & -14 \\ 0 & 27 & 0 \\ -75 & 36 & 69 \end{bmatrix}$

$f(A) = \begin{bmatrix} -74 & 27 & -45 \\ 0 & 243 & 0 \\ -225 & 108 & 195 \end{bmatrix}$

$\underline{Q_3}$ $A = \begin{bmatrix} 1+2i & 0 & 5 \\ 3 & 1-2i & 2 \\ -3 & 5 & 4 \end{bmatrix}$

$\|A\|_{m_1} = \max_j \sum_i |a_{ij}| = \overset{\max}{\{}\sqrt{5}+6, \sqrt{5}+5, 11\} = 11$

$\|A\|_{m_\infty} = \max_i \sum_j |a_{ij}| = \max\{\sqrt{5}+5, \sqrt{5}+5, 12\} = 12$

$\|A\|_{m_F} = \sqrt{\sum_{ij} |a_{ij}|^2} = \sqrt{98} = 7\sqrt{2}$

$\|A\|_m = \sum_{ij} |ij| = 26.47$

$\|A\|_1 = \max \sum_i |a_{ij}| = \max(\sqrt{5}+6, \sqrt{5}+5, 11) = 11$

$\underline{Q.4}$ $A = \begin{bmatrix} 2 & 4 \\ 4 & 10 \end{bmatrix}$ $cond_2(A)$ $cond_\infty(A)$

$A^TA = \begin{bmatrix} 20 & 48 \\ 48 & 116 \end{bmatrix}$

$\det(A^TA - \lambda I) = \begin{bmatrix} 20-\lambda & 48 \\ 48 & 116-\lambda \end{bmatrix} = 0$

$= (20-\lambda)(116-\lambda) - (48)^2 = 0 \Rightarrow \lambda^2 - 136 + 1600 = 0$

$\lambda = \frac{136 \pm 121}{2}$

$\lambda_1 = 128; \lambda_2 = 8$

$\sigma_{max} = \sqrt{\lambda_1} = 8\sqrt{2}$; $\sigma_{min} = \sqrt{\lambda_2} = 2\sqrt{2}$

$cond_2(A) = \frac{\sigma_{max}}{\sigma_{min}} = 4$

$cond_\infty(A)$ $A^{-1} = \begin{bmatrix} 2.5 & -1 \\ -1 & 0.5 \end{bmatrix}$ $\therefore \|A\|_\infty = \max\{2+4, 4+10\} = 14$

$\therefore \|A^{-1}\|_\infty = \max\{2.5+1, 1+0.5\} = 3.5$

$cond_\infty(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty = 14 \times 3.5 = 49$

Q:5 $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots$$

$$A^2 = A,\ A^3 = A,\ \ldots,\ A^K = A;\ K \geq 2 \qquad A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(At)^K = t^K A \text{ for } K \geq 2$$

$$e^{At} = I + A\left(t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots\right)$$

$$e^{At} = I + A(e^t - 1)$$

$$e^{At} = \begin{bmatrix} e^t & 0 & \\ 0 & 1 & e^t - 1 \\ 0 & 0 & e^t \end{bmatrix}$$

$$\cos(A) = I\ (\cos(1) - 1)A$$

$$= \begin{bmatrix} \cos(1) & 0 & 0 \\ 0 & 1 & \cos(1) - 1 \\ 0 & 0 & \cos(1) \end{bmatrix}$$

Q.6 $e^t = \begin{bmatrix} 1+t & t & -t \\ -2t & 1-2t & 2t \\ -t & -t & 1+t \end{bmatrix}$

$$e^{At} = I + At$$

$$e^t = I + At$$

$$A = \frac{e^t - I}{t}$$

$$e^t - I = \begin{bmatrix} t & t & -t \\ -2t & -2t & 2t \\ -t & -t & t \end{bmatrix}; \quad A = \begin{bmatrix} 1 & 1 & -1 \\ -2 & -2 & 2 \\ -1 & -1 & 1 \end{bmatrix}$$

Q:7 $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$; $A^T A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\sigma_1 = \sqrt{5}, \ \sigma_2 = 0 \quad \sigma_3 \ 0$

$\Sigma = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$(A^T A) v = \lambda v \Rightarrow v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$U = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$

$A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q.8 $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$A - \lambda I = \begin{vmatrix} 1-\lambda & 3 \\ 3 & 1-\lambda \end{vmatrix}$

$\lambda = (1-\lambda)^2 - 9 = \lambda^2 - 2\lambda - 8 = 0$

$\lambda = 4, -2$

$B - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix}$

$= \lambda^2 - 2\lambda = 0$

$\lambda = 2, 0$

$A \otimes B = (4 \times 2, 4 \times 0, -2 \times 2, -2 \times 0) = (8, 0, -4, 0)$